Nearest Neighbor Algorithm

Find a phoneme for test frame using the ~~Euclidean distance~~ Mahalanobis distance measure.

Training Data:

/a/:	2	1	/i/:	3	2	/u/:	5	3
/a/:	1	2	/i/:	1	3	/u/:	4	2

Test Data: 2 3

Ans:

For /a/ phoneme:

Mean calculation:

$$\mu_{1a} = \frac{2+1}{2} = 1.5$$

$$\mu_{2a} = \frac{1+2}{2} = 1.5$$

Covariance calculation:

$$\Sigma_{1,a} = \frac{(1.5-2)^2 + (1.5-1)^2}{2} = 0.25$$

$$\Sigma_{2,a} = \frac{(1.5-1)^2 + (1.5-2)^2}{2} = 0.25$$

~~Inverse~~ Covariance Matrix:

$$\Sigma_a = \begin{bmatrix} \Sigma_{1,a} & 0 \\ 0 & \Sigma_{2,a} \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Inverse covariance matrix:

$$\Sigma_a^{-1} = \begin{bmatrix} \frac{1}{0.25} & 0 \\ 0 & \frac{1}{0.25} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

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For /i/ phoneme:

Mean calculation:

$$\mu_{1,i} = \frac{3+1}{2} = 2$$

$$\mu_{2,i} = \frac{2+3}{2} = 2.5$$

Covariance calculation:

$$\Sigma_{1,i} = \frac{(2-3)^2 + (2-1)^2}{2} = 1$$

$$\Sigma_{2,i} = \frac{(2.5-2)^2 + (2.5-3)^2}{2} = 0.25$$

Diagonal covariance Matrix:

$$\Sigma_i = \begin{bmatrix} 1 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Inverse covariance Matrix:

$$\Sigma_i^{-1} = \begin{bmatrix} \frac{1}{1} & 0 \\ 0 & \frac{1}{0.25} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

For /u/ Phoneme

Mean calculation:

$$\mu_{1,u} = \frac{5+4}{2} = 4.5$$

$$\mu_{2,u} = \frac{3+2}{2} = 2.5$$

Covariance calculation:

$$\Sigma_{1,u} = \frac{(4.5-5)^2 + (4.5-4)^2}{2} = 0.25$$

$$\Sigma_{2,u} = \frac{(2.5-3)^2 + (2.5-2)^2}{2} = 0.25$$

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